IIT Madras
Data Science
& AI Lab

NutriVision

AI-POWERED FOOD ANALYSIS

AI-Powered Indian Food Nutrition Estimator

Team Contribution

Team Contributions
Name (Roll No.) — Contributions
Sahil Raj (23f3001764) — EDA, model training, hyperparameter tuning, evaluation, deployment, technical report
Sahil Sharma (23f1001232) — Dataset processing, model training, evaluation, deployment pipeline and code
Aman Mani Tiwari (22f3003055) — EDA, project reports and documentation, user guide
Tasneem Shahnawaz (22f2001653) — Nutrition mapping, model training, custom dataset, non-technical report, README
Samreen Fathima (22f3000089) — Dataset analysis, model training, documentation, developer guide, codebase

Project Objectives

1. Segement & classify multiple food items in a single photograph
2. Estimate real-world portion weight from a 2D photo without depth sensors
3. Map foods to a structured Indian nutrition database
4. For estimated food weight provide the corresponding nutrients
5. Deploy as a Gradio web application

Pipeline Evolution — Detection-First → Segmentation-First

OLD PIPELINE (Milestones 1–3)

1. **YOLO Detection**
   Bounding boxes for food items
2. **SAM-2 Segmentation**
   Prompted by YOLO boxes
3. **MLP Weight Estimation**
   Pixel ratio → grams regression
4. **CSV + Pinecone RAG**
   Nutrition lookup
5. **Gemini**
   Generic dietary suggestions

REDESIGNED →

NEW PIPELINE (Milestones 4–6)

1. **SAM3 Segmentation**
   Direct pixel masks — no YOLO needed
2. **Coin Scale Detection**
   ₹10 coin = 2.7cm calibration reference
3. **ConvNeXtV2-Tiny**
   90% accuracy — 80 Indian food classes
4. **PCA Geometry**
   Rotation-invariant width/height extraction
5. **LLM Volume + Density**
   Ollama-Qwen + hard JSON density enforcement
6. **Nutrition Scaling**
   Per-100g → actual weight · Nutrition.json

Datasets — Multi-Track Training Approach

KHANA Dataset (Classification — Final)

Total images:	131,819 validated
Classes:	80 Indian food categories
Purpose:	Classification

YOLO Detection Datasets (Milestones 1–3)

• **IndianFood10 (Roboflow)**	**11,547 imgs · 10 classes**
• **IndianFoodNet-30**	**5,446 imgs · 30 classes**
• **South Indian Food Det.**	**9,338 imgs · 31 classes**
• **Indian Food YOLOv5**	**~14,000 imgs · 81 classes**
• **TOTAL (merged, M3)**	**39,185 imgs · 118 classes**

Nutrition Database

310+ Indian dishes · per-100g
(Kaggle 2025 11 nutrients tracked)

Custom Weight Dataset

~41 overhead plate photos with kitchen scale labels
₹10 coin placed in each frame

EDA & Preprocessing — Khana Dataset for the New Pipeline

What We Found (EDA)

Dataset Used for Classification

- 131,819 images across 80 Indian food classes
- Split: 80% train / 10% val / 10% test (stratified)
- Average 164 images per class

Class Imbalance Discovered

- Biggest class: 788 images Smallest: 12 images
- Without fixing this, the model learns common foods well but fails on rarer ones

Visually Similar Classes (Hard for AI)

- Pizza variants — look identical, toppings differ slightly
- Phirni vs kheer — nearly the same in photos
- These accounted for most model mistakes

What We Did (Preprocessing)

Class Balancing — WeightedRandomSampler assigned higher sampling probability to underrepresented classes, preventing dominant classes from biasing training.

Normalisation — Pixel values normalised using ImageNet mean and std to align with the pretrained model's expected input distribution.

Resizing — Images resized to 176px to match ConvNeXtV2's native input.

Augmentation — Flips, random crops, and brightness/contrast adjustments were applied.

Model Architecture — Classification Backbones

ConvNeXtV2-Tiny (CHOSEN)

Parameters:	27.9M
Input Size:	176 × 176 px
Top-1 Accuracy:	**97% | F1: 0.92**

EfficientNetV2-S (Comparison)

Parameters:	20.3M (tf_efficientnetv2_s)
Input Size:	300 × 300 px
Top-1 Accuracy:	92.85% | F1: 0.9024

Baseline: MobileNetV2 (128×128, 10 epochs, CPU) → 72.06% accuracy · Used to validate the approach, then superseded

SAM3 Result & Prompts

SAM3 (Segment Anything Model 3) is prompted with text queries to detect and segment individual food items along with their containers from a single meal image.

Text Prompts Used:

- "food and its bowl" — segments food together with its serving bowl
- "food" — targets standalone food items (e.g., roti, bread)
- "drink and its glass" — detects beverages with their containers
- "coin" — detects the Rs 10 calibration coin for pixel-to-cm scaling

Key Observations from Result:

- Each detected region is overlaid with a colored mask and confidence score (0.71–0.91)
- Multiple food items on a thali are successfully isolated as separate segments
- Bounding boxes are generated per item for downstream PCA-based size estimation

Live Deployed Application — NutriVision Web App

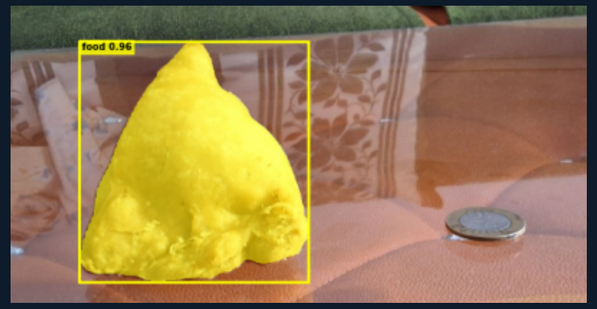
Primary URL (More Credits):

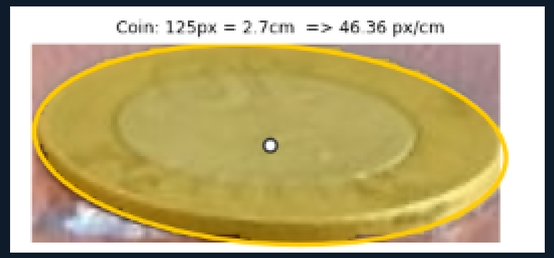
https://7868-01kpaaw75y1amk0fgq7x97120z.cloudspaces.litng.ai/

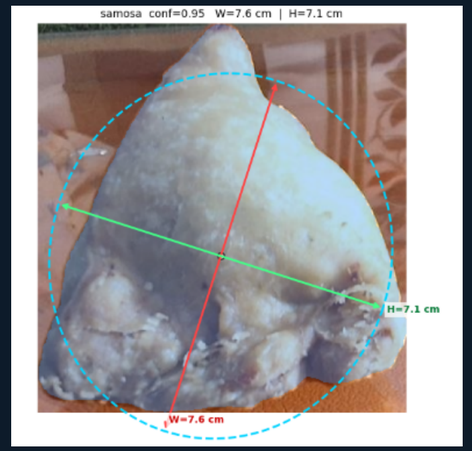
Old Web app URL:

https://7860-01knjbx5j5gnvmttvxykcmx4bh.cloudspaces.litng.ai/

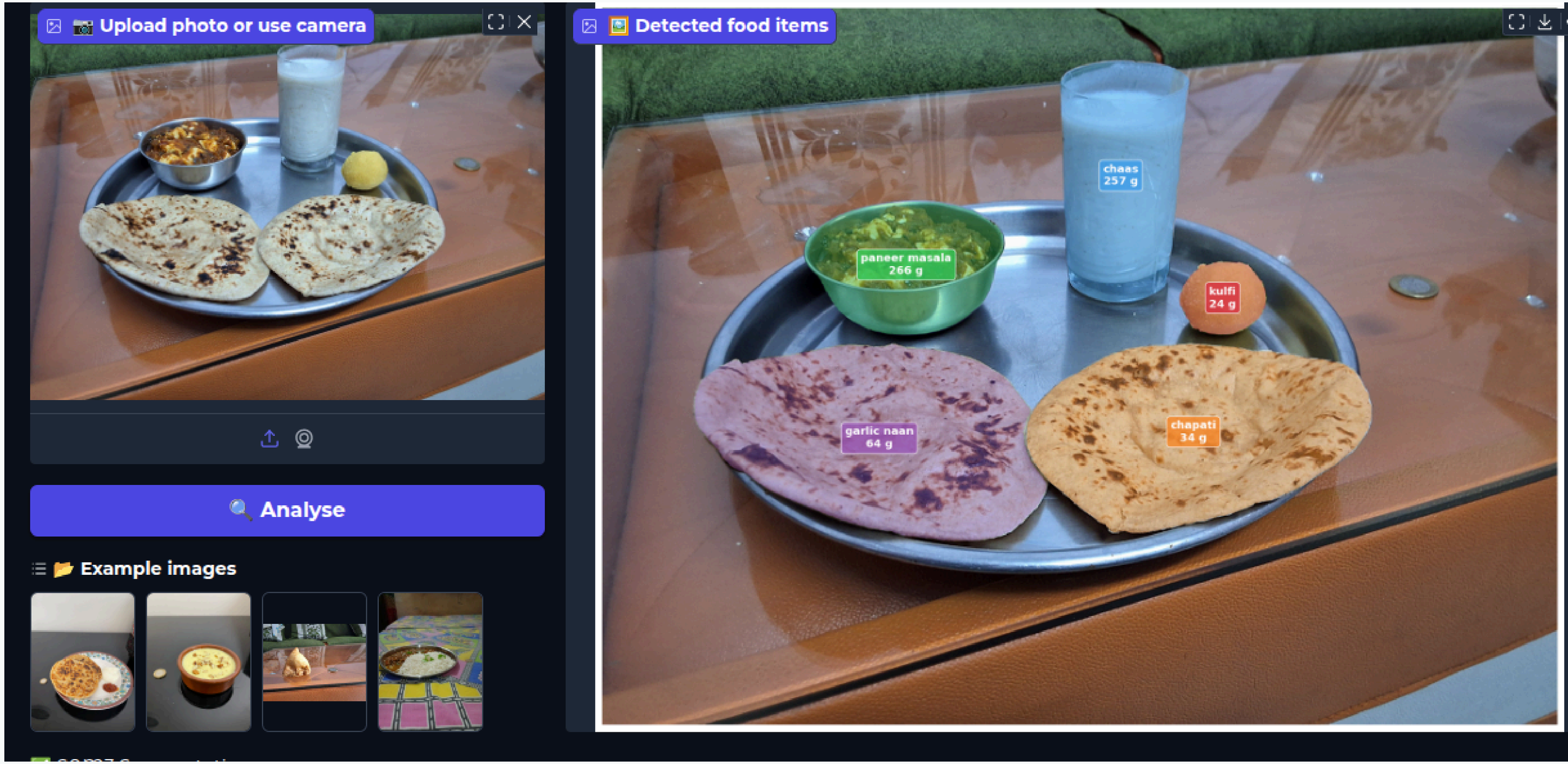
What happens while the app is analyzing your image in the background:







PCA is applied to SAM3 segmentation masks to extract major/minor axes (height & width) of each food item, with a Rs 10 coin (2.7 cm diameter) as the pixel-to-cm calibration reference. Real-world dimensions are then used to estimate volume via geometric approximation (cylinder or cuboid based on food category), and weight is derived as Volume × food-specific density from public references.



paneer masala->250gm
chass-> 280gm
chapati->40gm
ladoo->28gm

Key Findings & Limitations

Key Findings

- Top-5 accuracy 98% — correct class almost always in model's top 5
- Simple Prompting strategy is more efficient than YOLO multi-detection strategy
- PCA + coin calibration more practical than depth estimation

⚠ Limitations

⚠ Pizza-variant confusion (chicken/paneer/margherita) — visually irreducible in 2D

⚠ ₹10 coin must be fully visible & flat — pipeline halts without it

⚠ LLM latency: each food item = one API call (4 items ≈ 10 minutes total)

⚠ PCA weight estimate depends on clean segmentation & correct shape assumptions

⚠ 79 food classes only — unsupported dishes rejected

⚠ Dataset not cover dim lighting, partial occlusion, extreme angles

Future Improvements — Roadmap for Further Development

Classification

- Expand to 150+ food classes incl. regional specialties

USer's input for confirmation of classes and class for unclassified foods

Weight Estimation

- ARKit on iOS for LiDAR-based coin-free scale estimation

Data & Nutrition

- Cooking-method-aware values (fried vs. steamed biryani)
- Add Vit-D, B12, zinc, Mg micronutrients for vegetarian diets